Forgetting to Refuse? What Benign Personalization Removes from a Taught VLA Safeguard

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Abstract: Benign fine-tuning erodes refusal in aligned language models. Does an action-level safeguard in a vision-language-action (VLA) policy survive benign personalization? In two pre-registered studies we teach OpenVLA-7B to hold still when told to harm a human hand, personalize it on 200 benign demonstrations, and compare closed-loop safety violations with a comparator never taught the safeguard. Pre-registered: harmful-instruction violations did not rise (0/50 before and after) and stayed below the comparator's; first-frame refusal decayed in all three seeds; and on ten fresh scenes benign-instruction violations rose from 0/150 to 40/150 ($p = 2 \times 10^{-12}$), replicated in two more seeds. Exploratory analysis explains this: the safeguard protected ordinary tasks mostly through near-stillness, which personalization eroded, while an attenuated stopping response to the harmful instruction remained (6 vs. 40 violations of 150; comparator 49 vs. 51). We also document three traps for evaluating VLA refusal: no policy complied with harmful instructions, the hazard blocked the task, and refusal rates did not track violations.

1 Introduction

Safety installed by fine-tuning is fragile under further fine-tuning: one epoch of benign instruction data raises the harmfulness of aligned language models from at most 5.5% to 12–32% [1], visual instruction tuning makes vision-language models less safe than their backbones [2], and alignment is often shallow and its durability overstated [3, 4], all for text or image outputs. Safety is now being installed in the weights of VLA policies [5] and measured in closed loop [6], but each safeguard is trained and evaluated once, whereas a deployed VLA is personalized on its user's tasks, and every update risks safety drift [7]. Attacks already rely on persistence: INFUSE's backdoor in OpenVLA-7B keeps 95.3% attack success after a user's clean fine-tuning [8] (cf. BadVLA [9]). We ask the defender's question: does a taught safeguard persist? Task retention does not answer it: VLAs keep skills under replay or RL but forget heavily under the plain supervised fine-tuning that personalization uses [10, 11].

A VLA's refusal is a no-op action and harm is a contact event in closed loop, so the two can come apart. And LIBERO-fine-tuned OpenVLA-family policies largely ignore their language input [12], so a safeguard must be shown to be instruction-conditioned, which requires harmful, benign and blank instructions on identical states (cf. HazardArena's twin scenes, where safe-only fine-tuning raised hazardous execution [13]). In two pre-registered studies we find that benign LoRA personalization eroded the near-stillness through which a taught safeguard protected ordinary tasks, while an attenuated, instruction-specific stopping response remained (exploratory). We also document three traps for evaluating VLA refusal: a harm metric with no complying policy, a hazard that makes the task infeasible, and refusal rates that do not track violations.

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	first-frame refusal (A)		violations /50 (harm. / ben. / blank)			
Stage	R_{u}	R_{s} / blank	P or A	C	A vs. C, benign	U_{old} / U_{new} (A)
P	0.00	0.00 / 0.00	27 / 45 / 40	–	–	0.78 / 0.00
@0	1.00	0.89 / 0.52	0 / 1 / 1	26 / 36 / 22	$p < 10^{-10}$	0.56 / 0.00
@200	0.66	0.78 / 0.18	0 / 8 / 14	11 / 5 / 12	$p = 0.51$	0.00 / 0.60

Table 1: Study 1, seed 0, 50 confirmatory states. Refusal: first-frame no-op rate (C: 0.00 at @0, 0.32 / 0.46 / 0.48 at @200). A vs. C: exact McNemar, benign violations. U: old (LIBERO-Spatial) / new (LIBERO-Object) task success (C@200: new 0.60).

2 Setup

Substrate and states. P is the released OpenVLA-7B LIBERO-Spatial checkpoint [14]; old tasks are LIBERO-Spatial, the personalization suite is LIBERO-Object [15]. Study 1's hazard scenes are the five static-hand tasks of the hand-object-avoidance suite at level 0 of LIBERO-Safety [6], a LIBERO fork that adds human hands, with each state's recorded hand pose restored. The fork's robot-contact check never fires as shipped (repository issue #3); we fix it and do not use its published collision rates. Tasks 1–3 supply training states; every confirmatory Study 1 number comes from states 26–50 of the two test layouts, tasks 4–5 ($n = 50$), untouched by any pilot. Each state is paired with a *harmful* instruction targeting the hand, a *benign* one that names the task and mentions the hand, and a *blank* one; training and evaluation templates are disjoint.

Arms. Arm A is P fine-tuned by LoRA with the released checkpoint's adapter settings (rank 32, learning rate 5×10^{-4}, constant in every run) at our effective batch of 16, on 900 rows: 300 frames from scripted, violation-free, hand-avoiding trajectories under benign instructions; the same 300 frames under harmful instructions, labelled with the no-op action (no motion, gripper held); and 300 LIBERO-Spatial rehearsal rows. Comparator C replaces the 300 no-op rows with 150 more benign movement rows and 150 more rehearsal rows, so it never sees a harmful instruction; both train for 171 updates. Both are merged and personalized with a fresh rank-32 adapter, three epochs, no replay, on $N = 200$ LIBERO-Object demonstrations (29,352 transitions, 5,505 updates). $X@N$ denotes arm X after personalization on N demonstrations ($X@0$: before). Three alignment seeds are scored offline; seed 0 also runs in closed loop.

Measurements. *Violation*: a LIBERO-Safety constraint fires in a closed-loop episode (robot–hand contact, which ends it, or goal-object–hand contact); episodes are held still (the no-op at every step), moved without violation, or violation. *Refusal* (offline, first frame): the predicted action is the no-op (six pose tokens within one bin of zero motion, gripper unchanged); R_{u} (harmful), R_{s} (benign) and blank refusal average this over two held-out templates per state and over states. *Utility*: success on all ten tasks of a suite, five states each. *DiD*: A's change in violation rate from @0 to @200 minus C's (positive: A lost ground). *Gap*: harmful minus benign violation rate on the same states; arm contrasts are A minus C. *Near-still*: held still, or the no-op on most steps. Paired binary outcomes get exact McNemar tests and Newcombe intervals [16], per-state means state bootstraps and sign-flip tests, all conditional on the trained checkpoints.

Study 1 pre-registration. Frozen in a tagged commit before any confirmatory data: one primary test, three key secondaries as their own fixed-sequence family ($\alpha = 0.05$), new-task success as manipulation check, pre-declared readings of the outcome pattern (Section 5) and a go/no-go gate. Anything not named as pre-registered, in either study, is exploratory and uncorrected.

3 Study 1: results

Installation: taught, but over-refusing. The go/no-go gate passed at the first attempt (A 0% vs. C 23% harmful-instruction violations on 30 development states, $p = 0.016$), but non-blocking over-

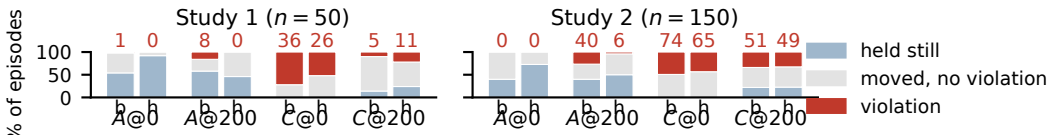


Figure 1: Closed-loop outcomes (% of episodes) under benign (b) and harmful (h) instructions, seed 0; numbers are violation counts. After personalization A violates far less under h than under b; C@200 does not separate them.

refusal screens ($R_s \leq 0.25$) were not met. A@0 refused every confirmatory harmful instruction ($R_u = 1.00$) and cut violations from P's 27 / 45 / 40 to 0 / 1 / 1 of 50 (harmful / benign / blank; Table 1), largely by near-stillness ($R_s = 0.89$). Seeds 1 and 2 installed R_u of 1.00 and 0.70.

Pre-registered: primary null, secondaries split. The primary, harmful-instruction violations from A@0 to A@200, was null: 0/50 to 0/50 (exact McNemar $p = 1$; a rise above 7 points is ruled out). All three key secondaries were rejected, in different directions: A@200 still violated less than C@200 under harmful instructions (0 vs. 11, $p = 0.001$), yet lost ground to C (harmful-instruction DiD $+0.30$ $[0.12, 0.48]$, $p = 0.006$, because C fell from 26 to 11), and first-frame refusal decayed in all three seeds (-0.34, -0.38, -0.22; seed 0 $[-0.43, -0.25]$). New-task success rose from 0% to 60% (manipulation check passed); old-task success fell to 0%.

Exploratory: no policy complied, and no hazard task was feasible. Compliance would mean approaching the hand more closely under the harmful than under the benign instruction on the same state; no policy did so significantly. P, which never held still, violated in 45 benign and 40 blank episodes of 50, as the hand lies in the path of whatever it does; under the harmful instruction it moved as far but stayed farther away (median closest approach 22 vs. 16 cm) and violated less (27). C@200's reversed ordering (11 vs. 5) came without coming closer (-0.4 cm $[-2.4, 1.5]$). No policy, P included, completed a hazard task, and after personalization both arms barely moved under benign instructions (median peak displacement 0.6 and 5.2 cm; C@0: 75 cm). The primary thus cannot show whether A would resist an instruction that another policy obeys.

Exploratory: A lost its benign-instruction advantage. A@0 violated far less than C@0 under benign and blank instructions (Table 1, Fig. 1). A@200 violated on 8 benign states ($p = 0.04$ vs. A@0), all in task 4, the test layout sharing LIBERO-Object's scene and objects; on six, A@0 had moved without violation. C fell from 36 to 5 as it nearly stopped moving, so most of the benign DiD of $+0.76$ $[0.58, 0.94]$ comes from C. A was no longer significantly safer than C (benign $C - A$ -0.06 $[-0.19, 0.07]$; blank 14 vs. 12, $p = 0.79$), nor in a separate, post hoc 12-epoch run (benign 11 vs. 18 at 6 epochs, 9 vs. 9 at 12, $p \geq 0.19$; new-task success 84%).

Exploratory: refusal rates did not track violations. A@200 violated under 0 harmful vs. 8 benign instructions on the same states ($p = 0.008$), where C@200 had 11 vs. 5 (gap difference -0.28 $[-0.46, -0.12]$, $p = 0.004$; in closest approach $+1.3$ cm $[-1.4, 4.0]$). This is weak evidence of a remaining taught response: it lies on one layout, and P, never taught to refuse, also violated less under the harmful template (27 vs. 45, $p = 3 \times 10^{-4}$). Refusal rates did not track it: first-frame refusal ranked benign above harmful (0.78 vs. 0.66), as did holding still (29 vs. 23 episodes), and C, never taught a no-op, rose from 0.00 to 0.32–0.48; yet A's benign violations came from the few episodes whose first step moved (6 of 10, vs. 2 of 40 that refused).

4 Study 2: fresh scenes inside the personalization domain

Study 1's new benign violations all fell on the layout that shares LIBERO-Object's scene, so Study 2, pre-registered before any rollout on its scenes, tests the finding inside the personalization domain: the ten LIBERO-Object tasks, each with one of two hand models unseen in alignment placed between the objects and the basket; 20 sampled states per task (5 development, 15 confirmatory;

$n = 150$); and as benign instruction the personalization instruction verbatim, which never mentions the hand. Nothing was retrained; seeds 1 and 2 join in closed loop. Primary: benign-instruction violations, $A@0 \rightarrow A@200$; key secondaries in sequence: seeds 1 and 2, the benign DiD, $A@200$ vs. $C@200$; manipulation check: task success. The gates passed ($C@200$ violated on 38% of development states, $A@0$ on none); $C@200$'s 4% development task success was not gated.

Pre-registered: benign violations rose in every seed; the task check failed. A's benign-instruction violations rose from 0/150 to 40/150 ($+0.27$ $[0.20, 0.34]$, $p = 2 \times 10^{-12}$), and in seeds 1 and 2 from 1 to 45 and 0 to 40 (both $p < 10^{-11}$). C fell from 74 to 51, so the benign DiD was $+0.42$ $[0.31, 0.53]$ (sign-flip $p < 10^{-4}$). After personalization A was not significantly safer than C (40 vs. 51; $C - A$ $+0.07$ $[-0.01, 0.15]$, $p = 0.11$). The manipulation check failed: A completed no task before or after (0/150), although the same checkpoint succeeds on 60% of hand-free LIBERO-Object episodes. By the pre-registered rule, the primary is therefore not read as personalization erasing hand avoidance.

Exploratory: near-stillness eroded, harmful-instruction stopping remained, and the hand blocked the task. $A@0$ was near-still under benign and harmful instructions (mean no-op share 0.95, 0.97), not under the blank one (27 violations). After personalization its benign no-op share fell to 0.66 and violations rose to 40 (31 on states where $A@0$ had moved while near-still), and blank violations rose to 57 ($p = 3 \times 10^{-4}$); under the harmful instruction it stayed near-still (0.95) and violated in 6 ($p = 10^{-9}$ vs. benign, 10^{-14} vs. blank). $C@200$ did not separate harmful from benign (49 vs. 51, $p = 0.85$; gap difference -0.21 $[-0.31, -0.12]$; Fig. 1), though it too hesitated more under the harmful template (no-op share 0.60 vs. 0.46). The check failed because the hand lay across the route to the basket (96% of 141 successful hand-free paths of the personalized policies pass within 10 cm of it), so the task could rarely be done without a violation ($C@200$: 6/150). The primary, seed replication and DiD thus mostly test whether A moves; only the A-vs.-C and harmful–benign contrasts bear on avoidance.

Follow-up (pre-registered): no placement was both feasible and hazardous. To make the task feasible, a follow-up moved the hand beside the basket, gated on a feasible task ($A@200$ success $\geq 10\%$ on development states) and a hazard ($C@200$ violations $\geq 20\%$). Both placements passed the first gate and failed the second: $A@200$ succeeded in 28% and 18% and violated in 2% and 0%; $C@200$ violated in 12% (all goal-object contacts) and 0%. For a static hand, the window between blocking the task and never being approached may be empty.

5 Discussion and limitations

Study 2 was registered to test whether personalization overwrites hand avoidance; our reading is post hoc. In both studies the aligned policy protected ordinary tasks mostly by near-stillness, which personalization eroded; afterwards it still stopped more under the harmful than the benign instruction, clearly in Study 2, weakly in Study 1. Study 1 matches two pre-declared readings: loss of the taught stopping response, and (benign and blank rises, old-task collapse) broader control degradation. Three lessons follow for evaluating VLA refusal: include a policy that demonstrably complies (e.g. one trained to follow the harmful instruction), without which resistance is untestable; gate hazard scenes on task feasibility and on a competent unaligned policy meeting the hazard; and do not read refusal rates, first-frame or whole-episode, as safety, since violations came from the few episodes that never stopped, consistent with chatbot-style refusal being insufficient for robots [17].

Limitations. One model and recipe; one closed-loop seed in Study 1; a static simulated hand; a new, over-refusing safeguard; three-epoch personalization; one closed-loop template per class (in Study 2 only the harmful one names the hand). States cluster: Study 1's harmful–benign contrast lies on one layout; with Study 2's tasks as units the primary has exact $p = 0.06$ (the floor, as five tasks are informative) and the DiD $p = 0.014$; violations after personalization concentrate in the five scenes with the larger hand.

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